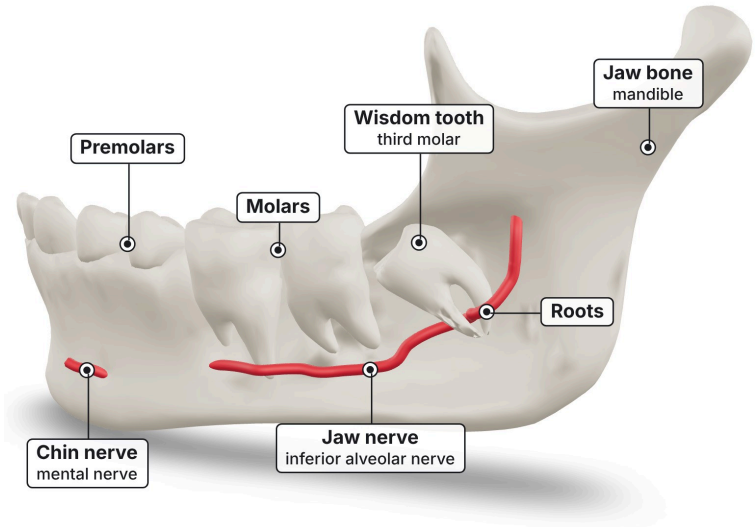
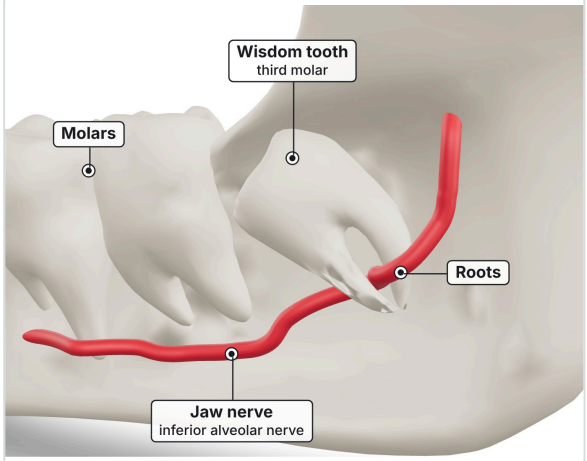
Wisdom tooth removal: the risk to your nerve

NAGOYA UNIVERSITY

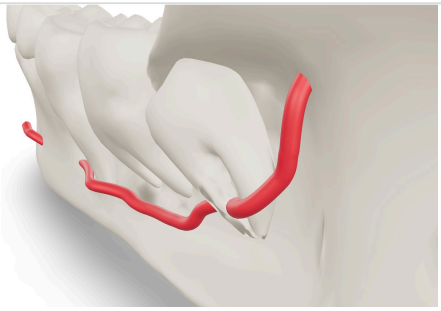
The roots of this wisdom tooth wrap around a nerve. Taking the tooth out can damage the nerve and make your lip and chin numb.



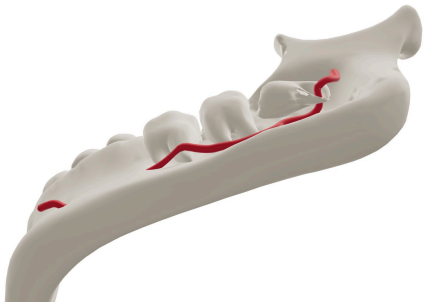
Whole jaw (cheek side)



Roots and nerve (close-up)



Roots around the nerve, from behind



The nerve in the jaw, from below

What is the problem?

Your wisdom teeth (**third molars**) are the last teeth at the back. An **impacted lower wisdom tooth** is stuck under the gum and bone. It has not come through the gum (it is **unerupted**). It cannot grow in the right way.

The nerve

A nerve runs in a tube of bone inside your lower jaw. Here we call it the **jaw nerve**. Its medical name is the **inferior alveolar nerve**. Two blood vessels run with it in the same tube. They are the **inferior alveolar artery and vein**.[1]

The jaw nerve gives feeling to your lower teeth and gums on the same side.[2] Near the front of your jaw, a branch of it comes out of the bone through a small hole. This branch is the **chin nerve** (**mental nerve**). It gives feeling to your lower lip and chin.[3]

Taking out a wisdom tooth can also hurt the nerve to your tongue (the **lingual nerve**).[4] About 1 to 3 in 1,000 people may get lasting damage.[5,6,7] This model does not show this nerve.

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

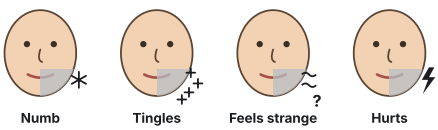
Why taking the tooth out can hurt the nerve

To take the tooth out, the surgeon must lift the roots out of the bone. The roots curve around the nerve. So the roots can press, stretch or cut the nerve as they come out. The artery and vein can also tear and bleed.[1]

The chin nerve branches off the jaw nerve. So damage to the jaw nerve can also stop the chin nerve working well.[3]

What nerve damage feels like

Your lower lip, chin, teeth or gums on that side may:[2,4]



For most people, the feeling comes back within weeks to months (**short-term** damage). For some, it does not fully come back (**long-term** damage).[2,8,9]

The older you are, the higher the risk, and the less likely the feeling is to come back fully.[2,10]

How often does it happen?

Some wisdom teeth have roots close to the nerve. Reviews of many studies looked at these teeth. Rates vary, but for these teeth:

About 7 in 100
may get nerve damage after the whole tooth is taken out[9,11,12,2]

About 1 in 100
may have nerve damage that does not go away[9,8,13]

If your dentist thinks the risk to the nerve is higher, they may refer you to a specialist (an oral surgeon) to take the tooth out.[14]

Sources **1.** Juodzbalys G, Wang HL, Sabalys G. J Oral Maxillofac Res. 2010;1(1):e2. doi.org/10.5037/jomr.2010.1102 **2.** Sarikov R, Juodzbalys G. J Oral Maxillofac Res. 2014;5(4):e1. doi.org/10.5037/jomr.2014.5401 **3.** Juodzbalys G, Wang HL, Sabalys G. J Oral Maxillofac Res. 2010;1(1):e3. doi.org/10.5037/jomr.2010.1103 **4.** Renton T, Yilmaz Z. J Orofac Pain. 2011;25(4):333-344. PMID 22247929. pubmed.ncbi.nlm.nih.gov/22247929 **5.** Lee J, Feng B, Park JS, et al. PLoS One. 2023;18(2):e0282185. doi.org/10.1371/journal.pone.0282185 **6.** Pichler JW, Beirne OR. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2001;91(4):395-401. doi.org/10.1067/moe.2001.114154 **7.** Petroni G, Passaretti A, Zaccheo F, et al. Eur J Dent. 2021;15(4):776-781. doi.org/10.1055/s-0041-1730042 **8.** Kang F, Sah MK, Fei G. J Stomatol Oral Maxillofac Surg. 2019;121(1):63-69. doi.org/10.1016/j.jormas.2019.06.010 **9.** Rico-Barroso L, Barroso-Medel M, Gutiérrez-Pérez J, et al. Med Oral Patol Oral Cir Bucal. 2026;31(1):e19-e30. doi.org/10.4317/medoral.27459 **10.** Queral-Godoy E, Valmaseda-Castellón E, Berini-Aytés L, et al. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2005;99(3):259-264. doi.org/10.1016/j.tripleo.2004.06.001 **11.** Derbishi AA, Altayyar RA, Alsubaiei AS, et al. Cureus. 2026;18(3):e105646. doi.org/10.7759/cureus.105646 **12.** Kang FW, Yuan XR, Li GC, et al. J Oral Maxillofac Surg. 2025;83(5):601-615. doi.org/10.1016/j.joms.2025.01.014 **13.** Cervera-Espert J, Pérez-Martínez S, Cervera-Ballester J, et al. Med Oral Patol Oral Cir Bucal. 2016;21(4):e505-e513. doi.org/10.4317/medoral.21074 **14.** Faculty of Dental Surgery, Royal College of Surgeons of England. Parameters of care for patients undergoing mandibular third molar surgery. London: RCS England; 2020. www.rcseng.ac.uk/-/media/Files/RCS/FDS/3rd-molar-guidelines--April-2021-v3.pdf

This sheet is for a research study at Nagoya University. Version 2.4, 21 October 2026.